PATHOPHYSIOLOGY — NURSING

Immune System — Part 2

NRSG 224 — Recorded lecture 5. Exam-detail notes from the class deck, her study guide, and the Patho sheet.

Same format as part 1. Slide and lecture content is black. **Red text is the add-on**: blanks on her Immune 2 study guide, plus what the full Patho study sheet fills in that the guide left empty. Sections are now in recording order, about 1 hour 37 minutes. Wording is unchanged. Teal times are where to be in the video.

1. Self-tolerance and MHC · 0:32

0:32 Self-tolerance is the absence of an immune response against a person's own antigens. Her outline left the "how" lines blank. The table on the same handout fills them.

- Suppressor / regulatory T cells suppress self-reactive immune cells and turn the response off.
- **3:09** Self-reactive T and B cells are eliminated during maturation. The Patho sheet names the sites: bone marrow screens B cells, thymus screens T cells.
- T cells do not activate unless the foreign antigen is complexed to the right MHC molecule. That stop is what keeps a helper T from firing by accident.

MHC — the display stand · 4:01

Major histocompatibility complex molecules sit on the cell surface and display antigens for T cells to inspect.

	Class I	Class II
Where	Nearly all nucleated cells. Mostly cells that do not phagocytize.	Mainly macrophages / APCs, which got the antigen by phagocytosis.
What they show	Intracellular: normal self antigens, viruses that got inside, abnormal proteins from mutation.	Extracellular antigens.
Who reads it	CD8 cytotoxic T. It only recognizes foreign antigen if it is on Class I.	CD4 helper T. No Class II display, no helper-T activation.

TEST TIP. Class I = inside the cell = CD8. Class II = outside, brought in by an APC = CD4. Same inside/outside split as part 1.

5:34 Practice short answer, her slide: why must a T cell see its antigen on the right MHC before it activates? So it cannot fire on an antigen that was never properly displayed. Class I gates the cytotoxic T. Class II gates the helper T. Without that match, adaptive immunity does not turn on against that target, including against self.

SHE SAID, NOT ON THE SLIDE. 15:49 Patho sheet only: HLA/DNA testing is how MHC type is checked. She noted it takes about 128 genes to code one MHC molecule, with 50–650 alleles. That is why a transplant match is hard.

21:05 Added from the recording: HLA is the human name for these MHC genes. That is the type they check for a transplant match, and it is a DNA test.

2. Autoimmunity · 21:58

Pathological change: breakdown of self-tolerance. Theories on her guide: abnormal self-antigen structure; faulty T-cell receptor / MHC interaction; disorder of immune surveillance; antigenic mimicry; sequestered antigens.

- **25:51** Antigenic mimicry: a foreign antigen looks so much like a self antigen that antibody made against it also hits self.

Practice short answer: why can group A beta-hemolytic strep set up an autoimmune disease? Antigenic mimicry. Those strep antigens look like heart and kidney. Antibody against the strep can then hit heart (rheumatic heart disease) or kidney (glomerulonephritis).

- **28:24** Sequestered antigens: self antigens that never met lymphocytes during early development. Injury later releases them, and the system treats them as foreign.
- **29:44** Predisposing factors: genes, a trigger event (virus, microorganism, chemical, release of sequestered antigen), and gender / hormones. More common in women.

TEST TIP. Highest-risk practice item: a female whose mother has SLE. Female plus a first-degree relative beats male, smoking, old age, or a high red-cell count.

32:46 Added from the recording: autoimmune disease remits and exacerbates. A quiet stretch is not a cure.

33:39 Added from the recording: organ-specific means the attack stays on one tissue. Systemic means the same mechanism hits many organs. SLE is her systemic example.

Assessment and labs · 34:55

Autoantibody tests: ANA (antinuclear antibody), anti-DNA, anti-CCP, antiphospholipid, rheumatoid factor. Inflammatory markers: CRP, fibrinogen, ESR. A positive ANA points at autoimmunity, not at infection.

36:34 Added from the recording: CRP and fibrinogen are acute-phase proteins the liver makes when cytokines say inflammation is on. ESR is how fast red cells settle; they settle faster when those proteins are up. All three are inflammation, not a specific disease.

Interventions · 41:54

- Immunomodulators: immunosuppressive drugs, monoclonal antibodies, corticosteroids.
- Symptom control with anti-inflammatories: NSAIDs (Motrin, ibuprofen, aspirin). Corticosteroids are synthetic cortisol and carry a lot of side effects.
- Vocab item on the list and not in either guide section: plasmapheresis, filtering antibodies out of plasma.

3. SLE — the worked example · 43:18

Systemic lupus erythematosus. Antibodies against many self antigens, including nuclear antigens such as DNA and RNA.

- Cell damage releases DNA.
- Anti-DNA antibodies form.
- Those antibodies meet DNA from damaged cells anywhere in the body.
- Antigen-antibody complexes deposit in tissues, activate complement, cause inflammation, cause more cell damage, and the cycle repeats. This is type III.

Assessment — blank on her guide

Bruising, rash, malar / butterfly rash, lupus arthritis, and involvement of kidney (renal failure), heart, eyes, lungs, GI, and blood cells. It remits and flares. Diagnosis is after other causes are ruled out.

Interventions — blank on her guide · 50:48

Same two levels as above: immunomodulators, and NSAIDs for symptoms. Corticosteroids are the drug she spends time on here, synthetic cortisol with a lot of side effects.

4. Inflammation · 54:07

54:07 Part 1 was immunodeficiency. Part 2 is what happens when the system aims at the wrong thing, or aims too hard.

Added from the recording: -itis means inflammation of that organ. Infection can cause it, but inflammation is not proof of infection.

Reaction of tissue to injury, regardless of cause. Purpose is to start healing: neutralize or destroy the harmful agent, limit its spread, and prepare damaged tissue for repair (clear dead tissue, bring nutrients).

Cardinal signs — the five blanks

Redness, heat, swelling, pain, impaired function.

Mechanism · 55:28

Mostly release of chemical mediators: histamine, prostaglandins, leukotrienes, plasma proteases, platelet-activating factor, other cytokines. Released from mast cells and immune cells.

- Vascular: brief vasoconstriction, then vasodilation. Capillary permeability rises, fluid escapes into the tissue.
- Cellular: white cells move into the injured area.
- Acute-phase, from cytokines: fever, more white cells, skeletal muscle catabolism.

SHE SAID, NOT ON THE SLIDE. Sheet fill: more blood, oxygen, fuel, and antibody in the area is why it is red and warm. The leak is why it swells, hurts, and loses function. First cells in are neutrophils. Monocytes and macrophages also arrive and can present antigen. Catabolism here means breaking down muscle protein. Fever is useful: many pathogens cannot take the heat, and some white cells work better.

Exudate and what goes wrong · 65:01

- Serous: clear, light yellow. Sero-sanguineous: watery blood. Purulent: pus.
- Problems if inflammation runs on: it gets out of control; chronic irritation can cause metaplasia; loss of function and deformity, with rheumatoid arthritis as her example.
- If the immune system is impaired, the opposite problem: the agent is not neutralized, spread is not limited, and repair does not get set up. That is the part 1 immunodeficiency picture.

TEST TIP. 65:01 Added from the recording: a fever the day after surgery is often the acute-phase response to tissue injury, not an automatic infection. Scar is the other cost she named when inflammation repairs by laying down collagen.

5. Hypersensitivity — ABCD · 74:17

Memory trick on the class slide: Allergies, antiBodies, immune Complex (Complement), Delayed. Types I, II, and III are antibody. Type IV is T cells. I–III are the immediate / B-cell side. IV is delayed.

Type	Name	What happens	Examples
I · 76:45	IgE / immediate	Allergen binds IgE already on mast cells or basophils. They degranulate and dump inflammatory mediators.	Allergic rhinitis (hay fever), asthma, urticaria, atopic dermatitis (eczema), anaphylaxis.
II · 90:28	Antibody-mediated / cytotoxic	IgG or IgM binds a cell or tissue antigen. The antigen-antibody reaction itself damages the target.	Blood transfusion reaction. Also myasthenia gravis: antibody against acetylcholine receptors, so muscle cannot contract.
III · 90:28	Immune complex / complement	Antigen-antibody complexes deposit in tissue, turn on complement, and inflammation does the damage. Not the binding itself.	SLE.
IV · 94:49	T-cell / delayed	T cells kill directly, or release cytokines that damage tissue. No antibody.	Contact dermatitis (poison ivy), diabetes mellitus. Sheet: type 1 diabetes, T cells on pancreatic cells. Also transplant rejection.

DO NOT MIX UP. Type II damage is the antigen-antibody reaction. Type III damage is the inflammation after the complex lands and complement fires. Same antibodies, different step does the harm.

DO NOT MIX UP. Allergy is a type of hypersensitivity, not a fourth immune system. Autoimmunity is the system attacking self. Immunodeficiency is the system failing to attack a real threat. AIDS is immunodeficiency, not autoimmune.

SHE SAID, NOT ON THE SLIDE. Type I extras from the sheet: angioedema is face swelling. Anaphylaxis is system-wide degranulation, blood pressure falls, and that is anaphylactic shock. Treatment side: antihistamines, anti-inflammatories, corticosteroids. Predisposing factors on the guide: a prior exposure that sensitized the person (inhaled, swallowed, injected, or on the skin), plus genetic makeup.

TEST TIP. 96:51 Added from the recording, her close: if it sounds like an allergy, it is Type I IgE. Hay fever, asthma, hives, eczema, anaphylaxis.

6. How she wants you to sort the disorders

Class slide 3 is the sort. These practice slides are not in the recording. They sit after Type IV.

Disorder	Problem	How you check	What you give
Immunodeficiency	Infection	Cultures, WBC count	Antimicrobials
Hypersensitivity	Allergic reaction. Unnecessary inflammation.	Allergy testing	Antihistamines and/or anti-inflammatory drugs
Autoimmunity	Damage to self tissue	Autoantibodies	Anti-inflammatory drugs, immunosuppressants

TEST TIP. All three can raise inflammatory markers. A high ESR or CRP does not tell you which box the patient is in.

7. Practice items from the class deck

Not in the recording. Class slides 7–12.

- Most reliable sign of infection: positive urine culture. Increased ESR is inflammation, not proof of infection. Increased WBC can be the acute-phase response. Positive ANA is autoimmunity.
- Highest autoimmune risk: a female whose mother has SLE.
- T cells and MHC: no display on the right class, no activation. That is the guard against firing on the wrong target.
- Group A beta-hemolytic strep: antigenic mimicry with heart and kidney.

8. Rapid-fire

- Immunodeficiency = infection. Hypersensitivity = unnecessary inflammation. Autoimmunity = self-tissue damage. All three raise inflammatory markers.
- ABCD: Allergies (IgE), antiBodies (cytotoxic), immune Complex (complement), Delayed (T cell).
- I–III antibody and immediate. IV is T cell and delayed. II damage is the binding. III damage is the inflammation.
- Class I / CD8 / intracellular. Class II / CD4 / extracellular. No Class II display, no helper T.
- Self-tolerance: suppressor T, delete self-reactive cells in marrow and thymus, MHC gate.
- Cardinal signs: redness, heat, swelling, pain, loss of function.
- Type I examples: hay fever, asthma, hives, eczema, anaphylaxis. Type II: transfusion, myasthenia gravis. Type III: SLE. Type IV: poison ivy, type 1 diabetes, transplant rejection.
- SLE cycle: DNA out, anti-DNA made, complexes deposit, complement, more damage. Butterfly rash. Rule-out diagnosis.
- Strep mimicry: heart and kidney. Culture beats ESR for infection. ANA is not an infection test.